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The Headline Says It All

The Problem

- It is tempting to think of data and data visualization as a neutral actor.
 - An emphasis on a minimalist aesthetic lends an air of objective, transparent reporting that masks visualization's persuasive power.
- Visualization are, in fact, rhetorical tools, built around facts.
- Given the growing ubiquity of visualization as a medium for recording, analyzing, and communicating data, we have a responsibility to examine how our design choices can influence the way a visualization is read, and what insights a reader walks away with.

Part 1: Select A Dataset

- The DEA Pain Pills Database (<https://www.washingtonpost.com/national/2019/07/18/how-download-use-dea-pain-pills-database/?arc404=true>). The Washington Post has published a significant portion of a database maintained by the Drug Enforcement Administration (DEA) that tracks every opioid from their manufacturer through to distributors and into pharmacies in towns and cities across the United States. This is an enormous database, and you can choose to work with it at any level of detail (e.g., state-wide, individual counties, or national summaries).
- Greenhouse Gas Emissions since 1990 (https://stats.oecd.org/Index.aspx?DataSetCode=AIR_GHG). The Organization for Economic Co-operation and Development (OECD) has compiled data for the emissions of all participating countries broken out by the pollutant (e.g., carbon monoxide, methane, etc.) and by different sources (e.g., energy, agriculture, etc.). The linked interface can be a little difficult to use, but you can access various slices of the data by either choosing alternate themes in the left-hand side menu or by customizing the pollutants and variables in the dropdown menus in the main view.
- Gender Equality Indicators since 1960 (<https://github.com/ZeningQu/World-Bank-Data-by-Indicators/tree/master/gender>). The World Bank tracks a number of different measures, including fertility rate, literacy, employment and ownership of businesses, and wages, to study the extent of gender equality around the world. The linked dataset curates a smaller subset of the [overall set](#) of gender indicators which you are welcome to use as well.
- Grab your favorite astronomy data...

Part 2: Understand the data

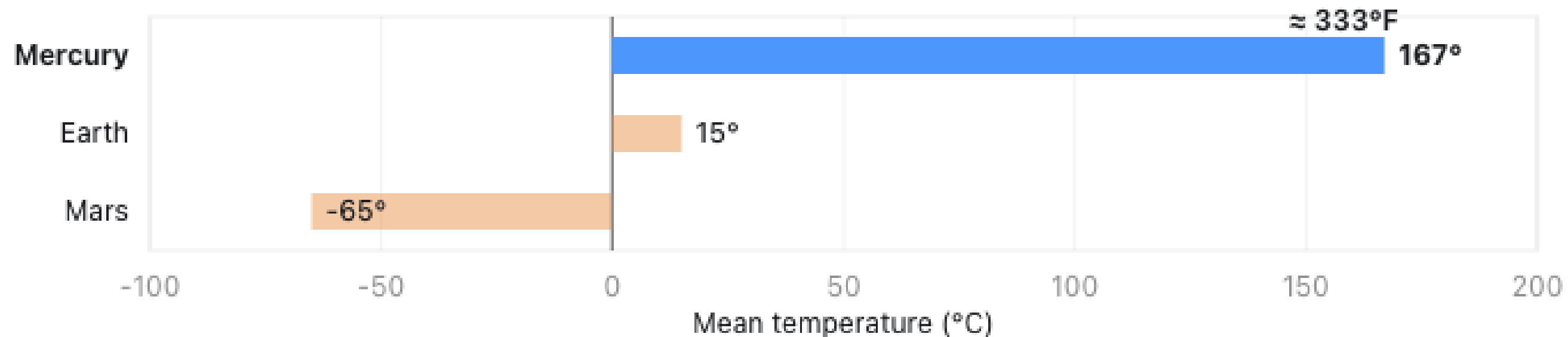
- Take some time to understand the data and explore it to better enable the next steps.

Part 3: Tell the story (version 1)

- Create a sensational headline that you can “support” with the data:
 - E.g., “Mercury is Hot!”
- Use your favorite LLM to create the best visualization you can using the data you’ve got to support that headline. You may need to iterate on your headline and visualization.

Mercury is hot—even averaged across the whole planet

Mean planetary temperatures, °C



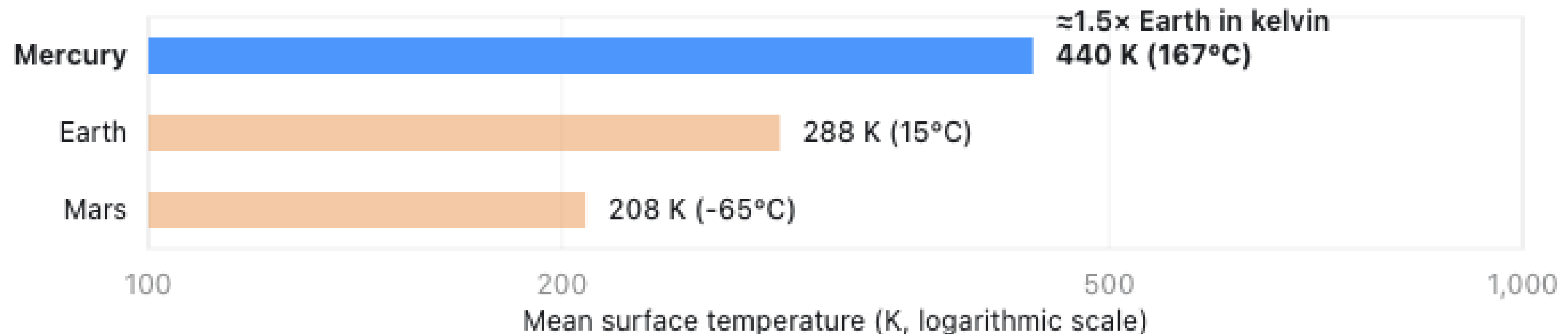
Source: NASA Planetary Fact Sheet, via NASA Science - Solar System Temperatures

Part 4: Tell the story (version 2)

- Create an alternative headline that is slightly less sensational:
 - E.g., “Mercury isn’t that hot!”
- Use your favorite LLM to build a visualization to support it. Again, you may need to iterate on this headline (or the first one).

Mercury isn’t that hot.

Mean surface temperature · kelvin, logarithmic scale



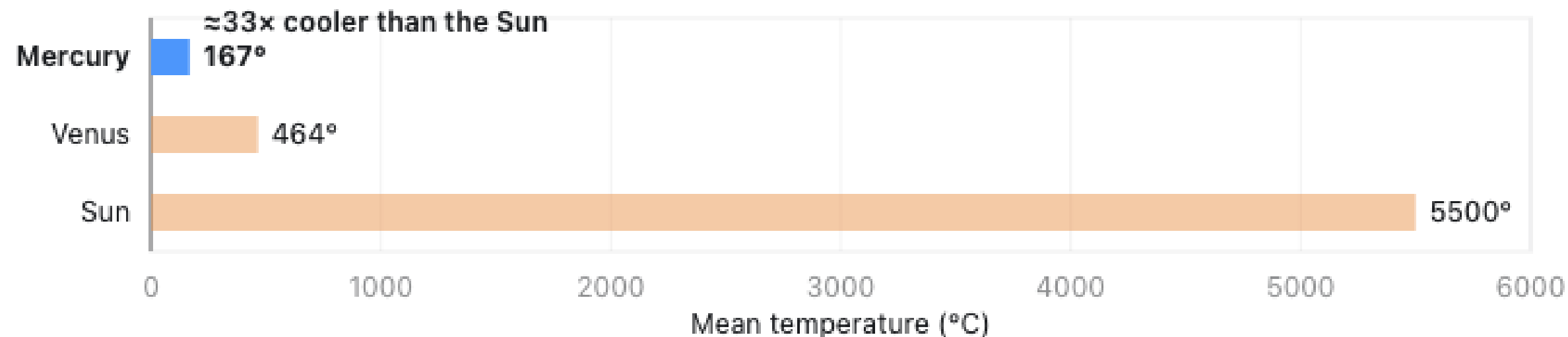
Source: NASA Science - Temperatures Across Our Solar System

Part 5: Tell the story (version 3)

- Create a completely contradictory headline:
 - E.g., “Mercury is cold!”
- Use your favorite LLM to build a visualization that supports it. You may again have to modify version 1 or 2.

Mercury is cold.

Compared with Venus and the Sun, °C



Source: NASA Science · Temperatures Across Our Solar System

Part 6: Present your visualizations

- Drop your visualization into the Google Drive
 - https://drive.google.com/drive/folders/1xGLbcbLfheiy-uKdP_6MBnBFFyCDWybo?usp=sharing
- Be prepared to describe what strategies you used to make your visualizations
- Be back at 5:30pm

Rules of the game and hints

- Work in groups of 2 or 3.
- All of your visualizations must use real data.
 - Cherry picking data is allowed!
- Use the tricks from the design lecture
 - Play with scales, encodings, etc. to tell your story.
 - Add “chart junk” to distract views.
 - Annotate your visualizations to focus attention.